

Problem Set 8—Suggested Answers

Exercises

Big picture. This problem set is about the payoff from imposing *distributional* assumptions on the error. So far in the course we derived the asymptotic normal distribution of $\hat{\beta}^{\text{OLS}}$ without assuming anything about the shape of $e_i|X_i$ beyond finite moments. The central limit theorem did the work and gave us an *approximate* normal distribution.

Once we assume $e_i|X_i \sim N(0, \sigma_e^2)$ exactly, we can derive the *exact* conditional distribution of $\hat{\beta}^{\text{OLS}}$ without any asymptotics. That exact distribution is the starting point for exact hypothesis tests—tests whose stated size is not “approximately 5%” but “5%, full stop.”

Model. $Y_i = \beta X_i + e_i$, where X_i, Y_i are scalars, and $e_i|X_i \sim N(0, \sigma_e^2)$ iid. There is no intercept. Define $\text{se}(\hat{\beta}^{\text{OLS}}) := \sqrt{\sigma_e^2 / \sum_{i=1}^N X_i^2}$.

(a) Derive the exact conditional distribution of $\hat{\beta}^{\text{OLS}}$.

Step 1: Write down the OLS estimator.

For the no-intercept model $Y_i = \beta X_i + e_i$, the OLS estimator solves $\min_b \sum_{i=1}^N (Y_i - bX_i)^2$, giving normal equation $\sum_{i=1}^N X_i(Y_i - \hat{\beta}^{\text{OLS}} X_i) = 0$, hence:

$$\hat{\beta}^{\text{OLS}} = \frac{\sum_{i=1}^N X_i Y_i}{\sum_{i=1}^N X_i^2}.$$

Step 2: Decompose.

Substitute $Y_i = \beta X_i + e_i$:

$$\hat{\beta}^{\text{OLS}} = \frac{\sum_{i=1}^N X_i(\beta X_i + e_i)}{\sum_{i=1}^N X_i^2} = \beta + \frac{\sum_{i=1}^N X_i e_i}{\sum_{i=1}^N X_i^2}.$$

The OLS estimator equals the true β plus a weighted average of the errors e_i , with weights $w_i := X_i / \sum_{i=1}^N X_i^2$. Crucially, *conditional on* $(X_1, \dots, X_N)$ these weights are non-random constants.

Step 3: Conditional mean.

$$\mathbf{E}(\hat{\beta}^{\text{OLS}} | X_1, \dots, X_N) = \beta + \frac{\sum_{i=1}^N X_i \mathbf{E}(e_i | X_i)}{\sum_{i=1}^N X_i^2} = \beta + 0 = \beta.$$

Step 4: Conditional variance.

Because $e_1, \dots, e_N$ are independent conditional on $(X_1, \dots, X_N)$:

$$\begin{aligned}\text{Var}(\hat{\beta}^{\text{OLS}} | X_1, \dots, X_N) &= \text{Var}\left(\frac{\sum_{i=1}^N X_i e_i}{\sum_{i=1}^N X_i^2} \mid X_1, \dots, X_N\right) \\&= \frac{1}{\left(\sum_{i=1}^N X_i^2\right)^2} \text{Var}\left(\sum_{i=1}^N X_i e_i \mid X_1, \dots, X_N\right) \\&= \frac{1}{\left(\sum_{i=1}^N X_i^2\right)^2} \sum_{i=1}^N \text{Var}(X_i e_i \mid X_i) \\&= \frac{1}{\left(\sum_{i=1}^N X_i^2\right)^2} \sum_{i=1}^N X_i^2 \sigma_e^2 = \frac{\sigma_e^2}{\sum_{i=1}^N X_i^2}.\end{aligned}$$

The step $\text{Var}(\sum_{i=1}^N X_i e_i | X_i) = \sum_{i=1}^N \text{Var}(X_i e_i | X_i)$ uses conditional independence of the e_i ; the step $\text{Var}(X_i e_i | X_i) = X_i^2 \sigma_e^2$ uses $e_i | X_i \sim N(0, \sigma_e^2)$.

Step 5: Normality.

$\hat{\beta}^{\text{OLS}} | (X_1, \dots, X_N)$ is a linear combination of the conditionally normal variables $e_i | X_i$. Any linear combination of independent normals is normal. Therefore:

$$\hat{\beta}^{\text{OLS}} | (X_1, \dots, X_N) \sim N\left(\beta, \frac{\sigma_e^2}{\sum_{i=1}^N X_i^2}\right).$$

Remark: exact vs. asymptotic. Without the normality assumption on $e_i | X_i$, the CLT would give us the *approximate* result $\hat{\beta}^{\text{OLS}} \stackrel{a}{\sim} N(\beta, \sigma_e^2 / \sum_{i=1}^N X_i^2)$ for large N . Here, normality of $e_i | X_i$ buys us the exact result for *any* N —a much stronger statement.

(b) Propose a sensible test of $H_0 : \beta = 0$ with exact size 5%.

What makes a test “sensible”? It should (i) have the stated size exactly, and (ii) have good power—it should actually detect departures from H_0 . The “silly” test in the problem sheet (reject if a random draw from $\{1, \dots, 20\}$ equals 8) has size exactly 5% but power also 5%; it is completely blind to the data.

Constructing the test. From part (a), under $H_0 : \beta = 0$:

$$\hat{\beta}^{\text{OLS}} | (X_1, \dots, X_N) \sim N\left(0, \frac{\sigma_e^2}{\sum_{i=1}^N X_i^2}\right).$$

Standardise:

$$T := \frac{\hat{\beta}^{\text{OLS}}}{\text{se}(\hat{\beta}^{\text{OLS}})} = \frac{\hat{\beta}^{\text{OLS}}}{\sqrt{\sigma_e^2 / \sum_{i=1}^N X_i^2}} \sim N(0, 1) \quad \text{under } H_0,$$

where the distribution is *exact* (conditional on $X_1, \dots, X_N$) and holds for any N .

The test. Reject $H_0 : \beta = 0$ if $|T| > 1.96$; do not reject if $|T| \leq 1.96$.

Size.

$$\mathbb{P}_{\beta=0}(|T| > 1.96) = \mathbb{P}(|Z| > 1.96) = 0.05, \quad Z \sim N(0, 1).$$

The size is exactly 5%, not just approximately.

Why is it sensible? The test uses the OLS estimator $\hat{\beta}^{\text{OLS}}$, which is a sufficient statistic for β in the normal linear model (it concentrates all the information about β in the data). Basing the test on $T = \hat{\beta}^{\text{OLS}}/\text{se}$ therefore uses the data as efficiently as possible. By the Neyman–Pearson lemma extended to composite hypotheses, this test is *uniformly most powerful unbiased* (UMPU): among all unbiased tests of size 5%, no other test has higher power at every $\beta \neq 0$.

Aside: the case of unknown σ_e^2 . If σ_e^2 is unknown, replace it with the consistent sample estimator $s_e^2 = \sum_{i=1}^N \hat{e}_i^2 / (N - 1)$, where $\hat{e}_i = Y_i - \hat{\beta}^{\text{OLS}} X_i$ and $N - 1$ reflects the one estimated parameter. Under normality of $e_i | X_i$, the resulting statistic $T^* = \hat{\beta}^{\text{OLS}} / \sqrt{s_e^2 / \sum_{i=1}^N X_i^2}$ has an exact t_{N-1} distribution under H_0 (by independence of $\hat{\beta}^{\text{OLS}}$ and s_e^2 in the normal model). For large N , the t_{N-1} quantiles converge to the normal quantiles, so the critical value 1.96 remains the same.

(c) **Derive and sketch the power function $p(\beta)$.**

Setup. We use the test from part (b): reject if $|T| > 1.96$ where $T = \hat{\beta}^{\text{OLS}}/\text{se}(\hat{\beta}^{\text{OLS}})$.

Under the *true* value β (not necessarily 0), part (a) gives:

$$\hat{\beta}^{\text{OLS}} | (X_1, \dots, X_N) \sim N\left(\beta, \frac{\sigma_e^2}{\sum_{i=1}^N X_i^2}\right),$$

so the test statistic has distribution:

$$T = \frac{\hat{\beta}^{\text{OLS}}}{\text{se}(\hat{\beta}^{\text{OLS}})} \sim N(\lambda, 1), \quad \text{where } \lambda := \frac{\beta}{\text{se}(\hat{\beta}^{\text{OLS}})} = \frac{\beta \sqrt{\sum_{i=1}^N X_i^2}}{\sigma_e}.$$

The quantity λ is the **non-centrality parameter**: it measures how far the true β is from zero in standard-error units. When $\beta = 0$ we have $\lambda = 0$ and $T \sim N(0, 1)$ (recovering part b). As $|\beta|$ grows, the distribution of T shifts away from zero and into the rejection region.

Derivation.

$$p(\beta) := \mathbb{P}_{\beta}(|T| > 1.96) = \mathbb{P}_{\beta}(T < -1.96) + \mathbb{P}_{\beta}(T > 1.96).$$

The trick is to centre T by subtracting λ , since $(T - \lambda) \sim N(0, 1)$:

$$\mathbb{P}_{\beta}(T < -1.96) = \mathbb{P}_{\beta}(T - \lambda < -1.96 - \lambda) = \Phi(-1.96 - \lambda),$$

$$\mathbb{P}_{\beta}(T > 1.96) = \mathbb{P}_{\beta}(T - \lambda > 1.96 - \lambda) = 1 - \Phi(1.96 - \lambda) = \Phi(\lambda - 1.96).$$

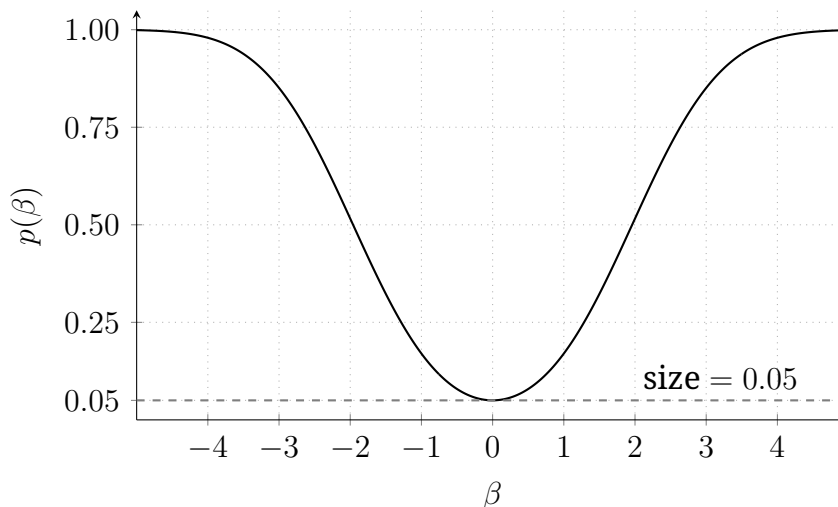
Therefore:

$$p(\beta) = \Phi\left(-1.96 - \frac{\beta}{\text{se}(\hat{\beta}^{\text{OLS}})}\right) + \Phi\left(\frac{\beta}{\text{se}(\hat{\beta}^{\text{OLS}})} - 1.96\right).$$

Verification of key properties.

- **Size at $\beta = 0$:** $p(0) = \Phi(-1.96) + \Phi(-1.96) = 2 \times 0.025 = 0.05$. ✓
- **Symmetry:** swapping $\beta \rightarrow -\beta$ swaps the two Φ terms, so $p(\beta) = p(-\beta)$. The power curve is symmetric around $\beta = 0$.
- **Consistency as $|\beta| \rightarrow \infty$:** $\lambda \rightarrow +\infty$ so $\Phi(-1.96 - \lambda) \rightarrow 0$ and $\Phi(\lambda - 1.96) \rightarrow 1$; hence $p(\beta) \rightarrow 1$. The test detects large effects with certainty.
- **Minimum:** $p(\beta) \geq p(0) = 0.05$ for all β ; the power never falls below the size. (A test where this holds is called *unbiased*.)

Sketch. Below is the power function for $\text{se}(\hat{\beta}^{\text{OLS}}) = 1$ (i.e., $\sum X_i^2 / \sigma_e^2 = 1$). The horizontal dashed line marks the size $p(0) = 0.05$.



Reading the power function.

- The curve has a “U” shape (sometimes called a “bathtub” shape) with minimum at $\beta = 0$.
- At $\beta = 0$ (on the null): $p(0) = 0.05$ is the probability of a type I error—we make a mistake 5% of the time even when we are testing the truth.
- At $\beta \neq 0$ (off the null): $1 - p(\beta)$ is the probability of a type II error. As $|\beta|$ grows the curve rises toward 1, meaning we make fewer type II errors.
- The rate at which $p(\beta)$ rises away from zero depends on $\text{se}(\hat{\beta}^{\text{OLS}})$. A larger sample (larger $\sum_{i=1}^N X_i^2$) shrinks se , which makes $|\lambda|$ larger for any fixed β , shifting the curve upward and giving the test more power. This formalises the intuition that larger samples make tests more powerful.

Remark: what the power function reveals about test design. The “silly” test from the problem sheet (reject if a random number equals 8, independent of the data) has power function $p(\beta) = 0.05$ for all β —a flat horizontal line at the size level. The test from part (b) has a power function that rises sharply away from zero precisely because it uses the data: $\hat{\beta}^{\text{OLS}}$ moves away from zero as $|\beta|$ grows, dragging T into the rejection region. The power function is therefore the right tool for evaluating and comparing tests.